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COS750 – Instructional Design Document

Forest of Keys

Preparing COS212 Students for Binary Search Trees

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Data structure: Binary Search Tree (BST)
Algorithm: BST Search

Summary

Forest of Keys is a browser-based learning application that prepares COS212 students for a lecture on binary search trees (BSTs). It focuses on one data structure, that is the binary search tree and one algorithm which is the BST search. The application teaches students how to identify the parts of a tree, recognise the BST ordering property, trace searches for keys that are present or absent, count comparisons, and explain how tree shape affects search effort.

A BST is a tree in which each node has at most two children. For distinct keys, smaller values are placed in the left subtree and larger values in the right subtree. During a search, each comparison determines whether the next step is to move left or right. This allows the search to discard the subtree that cannot contain the target. If the target is not found before the search reaches an empty position, the key is absent.

Scope and design decisions

The application assumes no prior knowledge of trees. Students move from labelled examples and demonstrations to partially completed activities, independent practice, and a final readiness check. The app gives explanatory feedback and suggests what to revise. Activities are ungraded, retries are available, and learning activity remains in the browser.

The algorithm is BST search, not a full treatment of all BST operations. BST insertion is used only in the sandbox to let students create different tree shapes and investigate their effect on search paths; insertion is not taught or assessed as a second algorithm. The design therefore removes the draft's separate insertion lesson, insertion mastery targets, delete-node functionality, and traversal simulator as core learning features. The sandbox may still use insertion as a tree-building interaction.

The design model is ADDIE. The instructional sequence uses gradual release of responsibility, supported by Gagné's nine events and Merrill's First Principles of Instruction. Learning activities also draw on constructivist, cognitive, behavioural, and social-cognitive principles where they directly support the stated outcomes.

Learning outcomes

Measurable learning outcomes

- After using Forest of Keys, a student will be able to:

LO1: Tree vocabulary

- Name and identify the root, parent, child, leaf, and subtree on a labelled tree diagram.

LO2: BST property

- State the BST ordering rule and decide whether a given tree satisfies it, including the ordering constraints throughout each subtree.

LO3: Search tracing

- Trace a BST search and list the nodes visited for both a key that is present and a key that is absent.

LO4: Comparison counting

- Predict the number of key comparisons required to search for a given key in a displayed tree.

LO5: Tree shape and efficiency

- Explain how balanced and lopsided tree shapes affect the number of comparisons, and explain why inserting keys in sorted order can produce a lopsided tree.

LO6: Readiness reflection

- Interpret the outcome-level results of the final check and select an appropriate revision route.

Instructional design model and learning theories

ADDIE:

- Analysis: The application is intended for COS212 students preparing for an introductory lecture on BSTs. It assumes no prior knowledge of tree terminology or BST search. The key learning need is to understand the structure, ordering rule, and search process before the lecture.
- Design: The six learning outcomes guide the lesson sequence, practice activities, feedback, and final check. Content is sequenced from vocabulary and the BST rule to search tracing, comparison counting, tree-shape exploration, and readiness reflection. Assessment items are mapped to the relevant outcomes.
- Development: The proposed web app contains labelled tree diagrams, step-by-step search animations, pseudocode, generated practice questions, a sandbox, and an outcome-level results screen. Explanations are available as text, with optional narration and visual demonstrations.
- Implementation: Students access the app in a browser and work through the activities at their own pace. The app provides hints, retries, and feedback. Nothing is marked, and learning activity remains in the browser.
- Evaluation: Diagnostic and formative checks identify misconceptions during learning. The final check reports performance by learning outcome and recommends a revision route. The design can be improved using student feedback and observed difficulty patterns.

Learning Theories Applied

Constructivism

- Students build their own BSTs by inserting values step by step.
- The BST is introduced as “binary search turned into a structure,” linking to prior knowledge of arrays and comparisons.
- Activities encourage prediction before explanation (e.g., “Where does 42 go?”).
- Learners discover the ordering rule themselves before it is formally stated.

Cognitivism / Information Processing

- Content is chunked into short lessons (3–5 minutes each).
- Dual coding: every rule is shown as text and an animated tree.
- Worked examples precede practice, reducing cognitive load.
- Signaling: current node highlighted, comparison shown alongside pseudocode

Behaviourism

- Immediate feedback: green highlight for correct, red with explanation for incorrect.
- Points, streaks, and badges reward progress.
- Repeated practice until mastery (e.g., three correct inserts in a row).

Social Cognitive Theory

- "Watch me" mode demonstrates search/insertion step by step.
- Students observe worked examples, then practice with guidance.
- Progress tracking (badges, progress bar) builds self-efficacy.
- Confidence grows through small successes and unlimited retries.

Catering to Different Learning Styles

Fleming's VARK Model

- Visual: Animated BST diagrams, colour-coded paths.
- Auditory: Optional narrated explanations.
- Read/Write: Pseudocode, text notes, downloadable summary sheet.
- Kinesthetic: Drag-and-drop node placement, interactive sandbox mode.

Kolb's Learning Cycle & Honey & Mumford Styles

- Concrete Experience (Activist) → Insert keys freely before rules are given.
- Reflective Observation (Reflector) → Prompts like "What do you notice about small vs large numbers?"
- Abstract Conceptualisation (Theorist) → Formal BST property and pseudocode revealed.
- Active Experimentation (Pragmatist) → Real-world tasks (e.g., build a BST of student IDs).

Motivation Strategies

Flow Theory

- Start simple (3-node BST), then scale up difficulty.
- Clear goals and immediate feedback maintain engagement.
- Adaptive difficulty ensures balance between challenge and skill.

Malone & Lepper's Framework

- Challenge: Levels increase in complexity; trick cases like invalid BSTs.

- Curiosity: Prompts like “What happens if we insert numbers in sorted order?”
- Control: Students choose pace, retry options, and sandbox values.
- Fantasy: Themed scenario (e.g., “Forest of Keys”) to make learning engaging.
- Recognition: Badges such as Tree Builder, Search Master, Traversal Explorer.

Keller’s ARCS Model

- Attention: Opening puzzle, animations, varied activities.
- Relevance: Links to real-world uses (databases, file systems).
- Confidence: Clear objectives, hints, unlimited retries.
- Satisfaction: XP, badges, progress bar, and a “Lecture-ready” certificate.

Assessment and Scaffolding

Diagnostic Check (Pre-Lesson)

- Short pre-test on comparisons, binary search, and linked-list nodes.
- Weak students shown refresher content before continuing.

In-Lesson Checks

- After each lesson, 2–3 questions with instant feedback.
- Two wrong answers unlock hints and worked examples.

Final Quiz

- 8–10 mixed questions tagged to Bloom’s levels:
    - Remember: “Which node is the root?”
    - Understand: “This tree is a BST. True or false?”
    - Apply: Drag-and-drop insert exercise.
    - Analyse: Compare two trees and explain why one is taller.

Scaffolding Rules

- 80%+: Lecture-ready, certificate awarded.
- 50–79%: Review weak topics, short re-test.
- Below 50%: Guided walkthrough with hints, then re-test.

User Journey

Screen 0: *Welcome*

- Hook: “How does your phone find a contact among 1000 in a split second?”
- Learning outcomes listed, reassurance: nothing is marked, retries are free
- Purpose: Introduce the application, set expectations and reduce anxiety. The learning outcome are briefly introduced in plain language.

Screen 1: *Introductory lesson*

- "What is a tree?"
- Views a labelled tree and selects terms to see what they mean
- Purpose: teach the basic vocabulary for LO1

Screen 2: *The BST rule*

- Learns that smaller values belong in the left subtree and larger values in the right subtree.
- Predicts where values belong and compares valid and invalid trees.
- Purpose: Develop understanding of the BST property (LO2), including the fact that it applies throughout a subtree, not only to direct children.

Screen 3: *How a search works*

- Uses a familiar higher-or-lower guessing analogy before following the search rules and iterative pseudocode.
- Purpose: Introduce the search process that students will apply when tracing a search (prepares for LO3).

Screen 4: *Worked example*

- Watches the application search for a key one step at a time. Each comparison is explained, the current node is highlighted, and an empty position is shown when the key is absent.
- Purpose: Demonstrate the search process and comparison counting (LO3 and LO4).

Screen 5: *Your turn*

- Completes partially worked search problems. Initially, the application provides some of the steps.
- The student gradually takes responsibility for more decisions until they can trace the full search independently.
- Purpose: Practice tracing searches for present and absent keys with decreasing support (LO3 and LO4).

Screen 6: *Practice*

- Answers mixed generated questions without step-by-step hints and receives explanatory feedback.
- The difficulty can change in response to performance.
- Purpose: Provide independent practice across the main content outcomes (LO2–LO5).

Screen 7: *Sandbox*

- Enters keys to build a tree and explores the search paths. The student can try inserting 1, 2, 3, 4, 5 and compare the resulting lopsided tree with a more balanced example
- Purpose: Explore how tree shape affects search comparisons (LO5). Insertion is used only to create trees for exploration, not taught as a second algorithm.

Screen 8: *Final Check*

- Answers 10 questions selected to sample the five content outcomes, LO1–LO5.
- Purpose: Check the student's current understanding of the content outcomes.

Screen 9: *Results*

- Views an overall result, an outcome-level report and a suggested revision route. An optional summary is available.
- Purpose: Support the student's reflection on readiness and choice of what to revise (LO6).

App Features Summary

- Interactive BST Builder: Insert/delete nodes, immediate feedback.
- Traversal Simulator: Animations, drag-and-drop exercises.
- Adaptive Quizzes: Scaled difficulty, timed challenges.
- Gamification Layer: Badges, XP, progress bar.
- Progress Dashboard: Tracks completed tasks, shows improvement.